

Student Identification

Name:

Class:

Date:

Objective

To quantitatively verify the Transformer Law ($\frac{V_s}{V_p} = \frac{N_s}{N_p}$) and analyze the conservation of energy and power efficiency in step-up and step-down configurations.

Theoretical Framework

A transformer works on the principle of **Faraday's Law of Induction**. The changing magnetic flux created by the primary AC voltage induces an electromotive force (EMF) in the secondary coil.

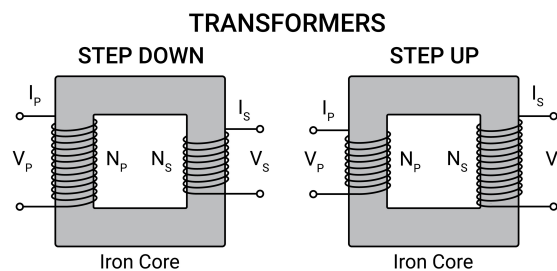


Figure 1: Transformer Schematic showing N_p , N_s , and the ferromagnetic core.

Experimental Data Collection

Trial	V_p (V)	N_p	N_s	Simulated V_s (V)	Theor. V_s ($V_p \cdot \frac{N_s}{N_p}$)
1	100	200	400	200	200
2	300	500	700	420	420

Elite Analysis: Power, Energy & Efficiency

In an *ideal* transformer, $P_p = P_s$. In a real transformer, energy is lost primarily due to Joule heating and eddy currents.

Advanced Problem: Set $V_p = 220V$. Adjust turns to achieve $V_s = 110V$. If the measured $I_p = 0.5A$ and $I_s = 0.9A$:

- Calculate Input Power (P_{in}): 110 W
- Calculate Output Power (P_{out}): 99 W
- Determine Efficiency ($\eta = \frac{P_{out}}{P_{in}} \times 100\%$): 90 %
- Energy lost as heat over 5 minutes ($E_{loss} = P_{loss} \times t$): 3300 J

Critical Thinking & Conclusion

Explain why the efficiency calculated above is less than 100% based on the physics of the magnetic core

According to the law of conservation of energy, energy cannot be created or destroyed but will change forms. Electrical energy is changed to heat energy, and there are 2 reasons why its lost in heat: 1. Resistance
2. Magnetic movement

shown in Figure 1.